



Agentic Repository Automation

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May 2026



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GitHub Next

Research, Build, Balance, Ship, Learn

GitHub Next

GitHub Next investigates the future of software development.

We are a team of researchers and engineers at GitHub, exploring things beyond the adjacent possible. We prototype tools and technologies that will change our craft. We identify new approaches to building healthy, productive software engineering teams.



We live in the most incredible of times

- **AI** and **traditional dev tools** make us all magicians when it comes to code
- Everything is changing, yet continuity is there too



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Continuous AI



Continuous AI

implemented by

GitHub Agentic Workflows



GitHub Agentic Workflows

=

The coding agents you love running automated in safe sandboxes with strong guardrails in GitHub Actions



GitHub Agentic Workflows

Agentic human processes powered GitHub.

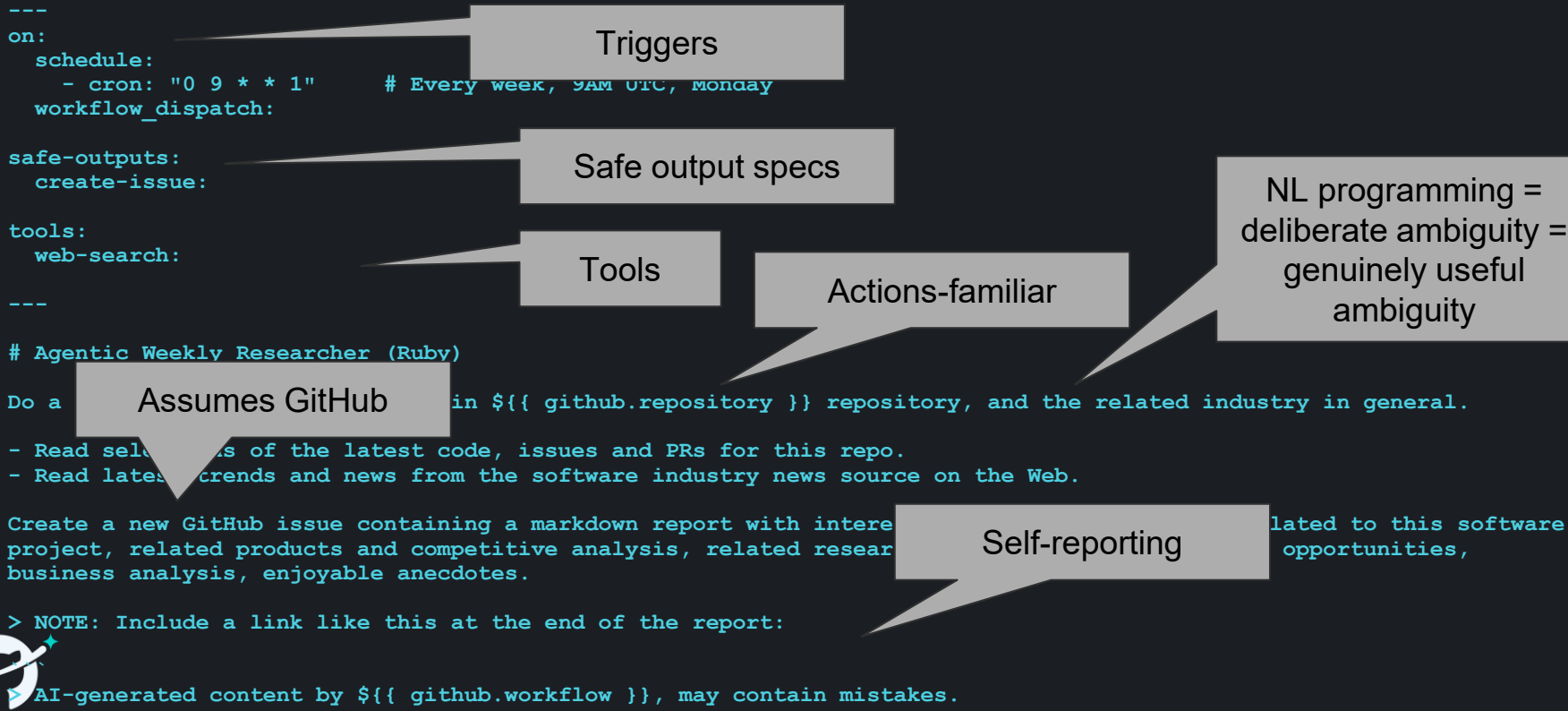


Workflows executed via GitHub Actions written in natural language.

Isolated sandboxes with scoped permissions and network control.

Agents interpret intent & operate on context across models. Copilot CLI · Claude · Codex · Gemini

Example - Weekly Researcher



Example - Issue Triage

```
---
on:
  issues:
    types: [opened, reopened]
```

```
safe-outputs:
  add-comment:
```

```
---
```

```
# Agentic Issue Triage
```

Your task is to analyze issue `\${{ github.event.issue.number }}` and perform some initial triage tasks related to that issue.

Analyze the issue content. Write notes, ideas, nudges, resource links, debugging strategies and/or reproduction steps for the team to consider relevant to the issue. Select appropriate labels already used in the repository and apply them to the issue.

Finalize an output report with your analysis and add it as an issue comment to the issue.

> NOTE: Include a link like this at the end of your issue comment:

```
```markdown
```

```
> AI-generated content by `${{ github.workflow }}`, may contain mistakes.
```

```
```
```



Agentic Workflows - Getting Started

```
gh extension install github/gh-aw
```

```
gh aw add-wizard githubnext/agentics/test-improver
```



Agentic Workflows - Safety, Safety, Safety

1. Execution is VM-sandboxed (GitHub Actions VMs)
2. Overall permissions are minimized
3. Agentic steps run essentially read-only
4. All outputs are "in the intended channel" ("safe outputs")
 - a. blast radius greatly reduced
 - b. checked by AI for safety
 - c. human oversight guaranteed
5. Outgoing network is firewalled and controlled



Continuous AI
implemented by
Agentic Workflows
enables
Agent Zoo



Continuous AI
implemented by
Agentic Workflows
enables
Continuous Test Improvement



Continuous AI
implemented by
Agentic Workflows
enables
Continuous Perf Engineering



Continuous AI
implemented by
Agentic Workflows
enables
Continuous Repository Maintenance



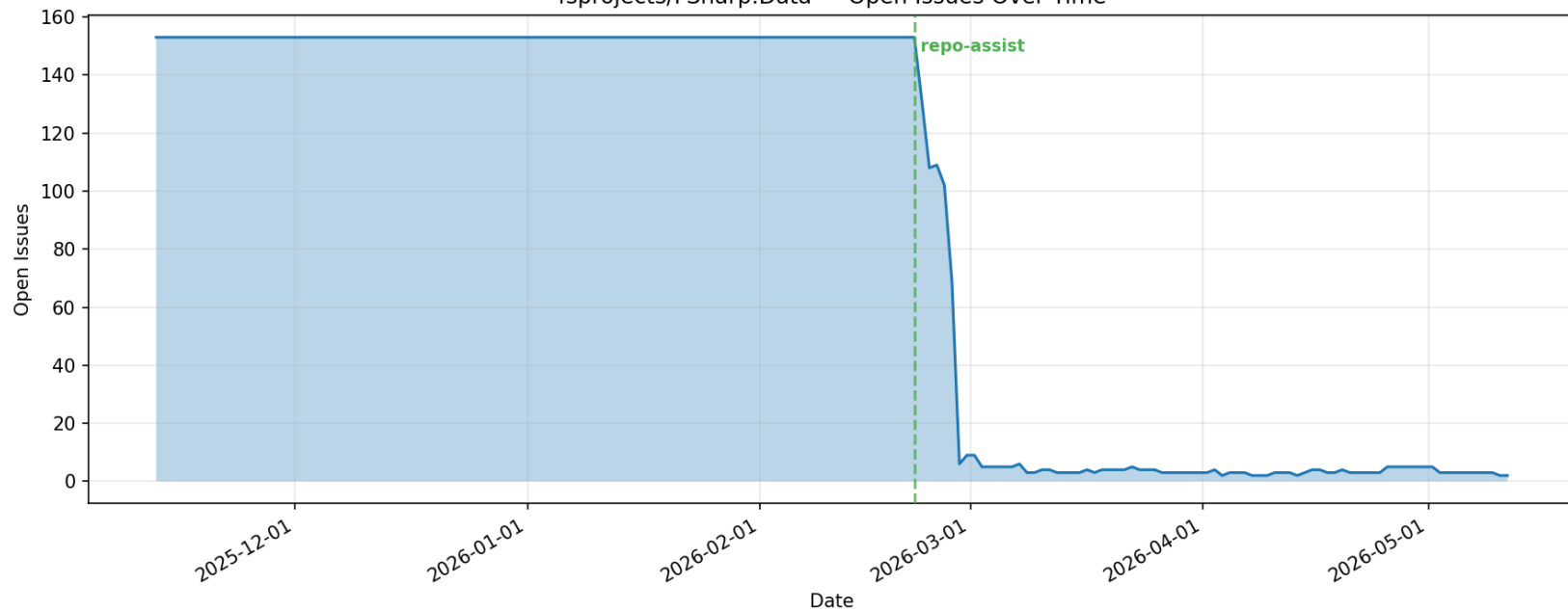
Continuous AI
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Agentic Workflows
enables
Continuous Formal Verification

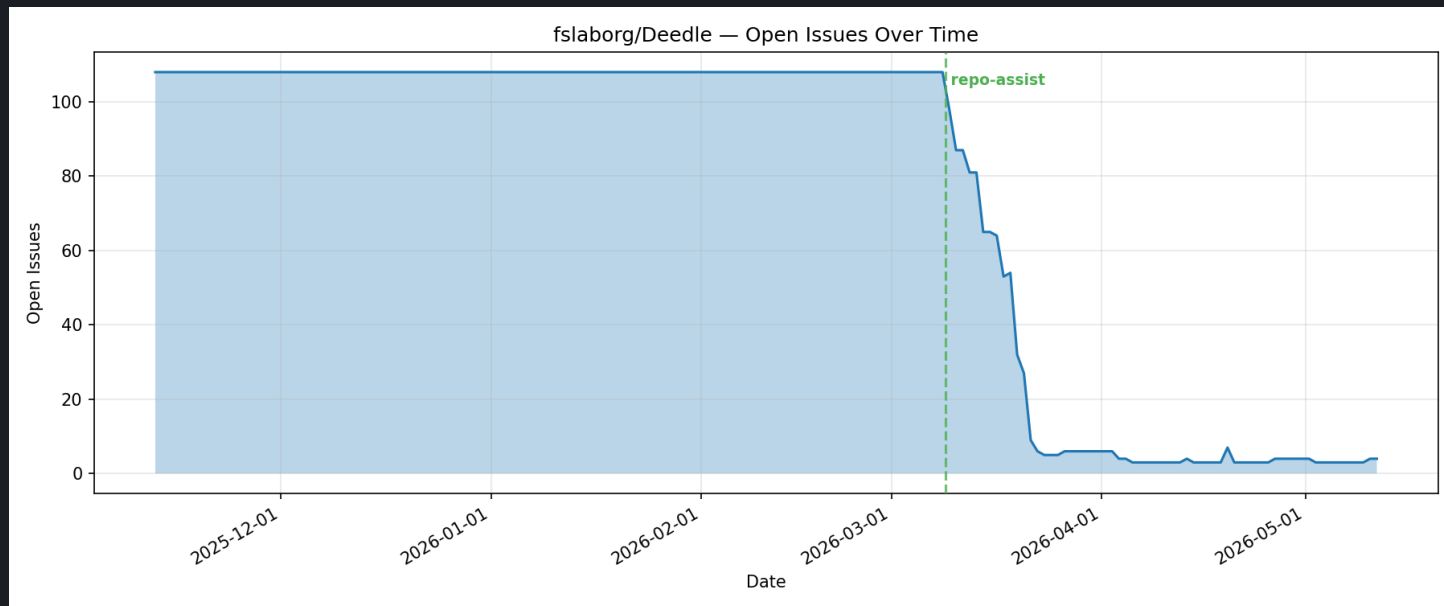


Demo – Repo Assist

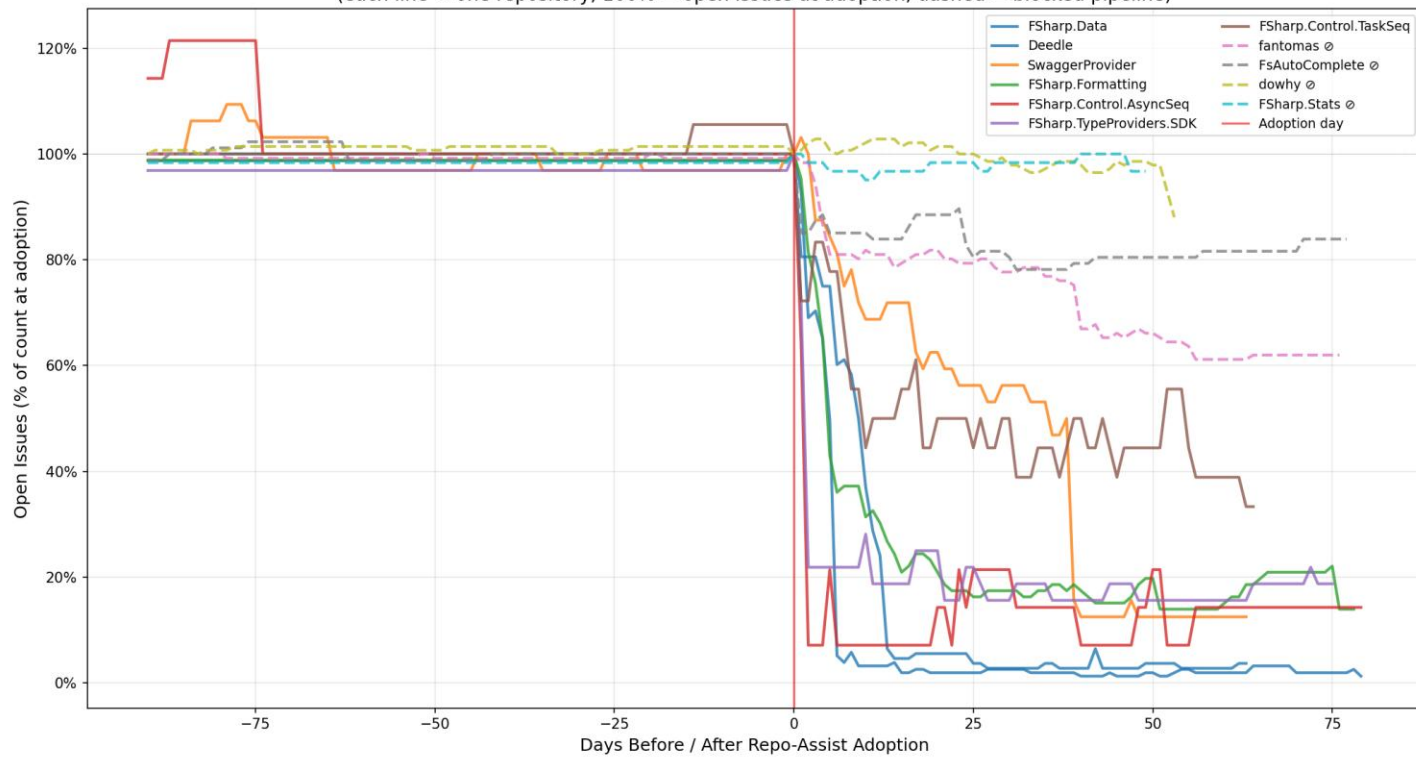


fsprojects/FSharp.Data — Open Issues Over Time





Open Issue Trajectories Normalized to Adoption Date
(each line = one repository, 100% = open issues at adoption, dashed = blocked pipeline)



Repository Throughput Analysis

| Repository | Comment Path (closed/total) | RA PRs | Merged | Rejected | Open (WIP) | PR Throughput | Status |
|-------------------|-----------------------------|--------|--------|----------|------------|---------------|---------|
| FSharp.Stats | 2/28 | 18 | 2 | 0 | 16 | 11% | BLOCKED |
| dowhy | 16/82 | 61 | 13 | 5 | 43 | 21% | BLOCKED |
| fantomas | 28/75 | 64 | 22 | 41 | 1 | 34% | BLOCKED |
| FsAutoComplete | 27/89 | 54 | 33 | 7 | 14 | 61% | BLOCKED |
| TaskSeq | 12/17 | 83 | 66 | 17 | 0 | 80% | FLOWING |
| FSharp.Formatting | 53/65 | 118 | 94 | 22 | 2 | 80% | FLOWING |
| TypeProviders.SDK | 11/16 | 50 | 45 | 4 | 1 | 90% | FLOWING |
| AsyncSeq | 7/7 | 69 | 56 | 11 | 2 | 81% | IDLE |
| FSharp.Data | 108/110 | 102 | 86 | 15 | 1 | 84% | IDLE |
| Deedle | 67/70 | 100 | 91 | 8 | 1 | 91% | IDLE |
| SwaggerProvider | 26/28 | 69 | 63 | 6 | 0 | 91% | IDLE |



Demo – Lean Squad



Theme 5: The Repository as an Automated Software Factory

Continuous Integration

Continuous Deployment

Continuous AI



Theme 5: The Repository as an Automated Formal Verification Factory

Continuous Integration

Continuous Deployment

Continuous AI





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software development

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AI Dev Tools are changing fast

- **2022: GitHub Copilot Code Completions** launched.
 - 1.8M+ paying users, 50%+ of users' code is written by Copilot
- **2023: AI Assists expand in traditional Developer Environments**
 - Summarization, advice, code review, e.g. Cody and others
- **2024: Shift to Task-orientation**
 - **GitHub Copilot Workspace, Cursor Composer, Devin** bring task-orientation to the IDE
- **2025: Coding Agents developed to explore and use tools**
 - Cursor/VS Code **Agent-Mode**, **Claude Code** and others launched
- **2025: AI-assisted app development** introduced
 - GitHub Spark, Bolt, Gemini, Lovable, Cogna, ...
- **2025: Continuous AI** introduced
 - GitHub Agentic Workflows, Google Jules, Curor Automations
 - Proactive, automated use of AI



Theme 6: New Futures, New Work

The Future is Work that wasn't being done before

Automobile

- Led to travel that wasn't being done before

Factory

- Led to millions of products that weren't being produced before

Medicine

- Led to many diseases that weren't being treated before

More overall work, not less

New work, not less work



Theme 6: New Futures, New Work

Software Performance Engineering was previously expensive, adhoc, ephemeral

- AI-automated performance engineering now tractable
- Needs human expertise and oversight
- **Good work that was too expensive to do**

Formal Verification was previously very expensive, rare

- AI-automated formal verification now looking tractable
- 100x less expensive
- Needs human expertise and oversight

Software Maintenance was previously very expensive, needed experts

- AI-assisted software maintenance now tractable
- 20x(?) less expensive
- Needs human expertise and oversight

